

Sulforaphane can protect lens cells against oxidative stress: implications for cataract prevention.



PURPOSE: Protecting the lens against oxidative stress is of great importance in delaying the onset of cataract. Isothiocyanates, such as sulforaphane (SFN), are proposed to provide cytoprotection against oxidative stress. We therefore tested the ability of SFN to perform this role in lens cells and establish its ability to delay the onset of cataract. METHODS: The human lens epithelial cell line FHL124 and whole porcine lens culture systems were used. The ApoTox-Glo Triplex Assay was used to assess FHL124 cell survival, cytotoxicity, and apoptosis. The MTS assay was used to assess cell populations. To determine levels of DNA strand breaks, the alkaline comet assay was performed and quantified. Lactate dehydrogenase levels in the medium were evaluated to reflect cell damage/death. To assess level of gene expression, an Illumina whole-genome HT-12 v4 beadchip was used. Protein expression was determined by Western blot and immunocytochemistry. RESULTS: Exposures of 30 μ M H₂O₂ to FHL124 cells caused a reduction in cell viability and increased cytotoxicity/apoptosis; these effects were significantly inhibited by 24-hour pretreatment with 1 μ M SFN. In addition, 1 μ M SFN significantly reduced H₂O₂-induced DNA strand breaks. When applied to cultured porcine lenses, SFN protected against H₂O₂-induced opacification. Illumina whole-genome HT-12 v4 beadchip microarray data revealed eight genes upregulated following 24-hour exposure to 1- and 2- μ M SFN, which included NQO1 and TXNRD1. This pattern was confirmed at the protein level. Nrf2 translocated to the nucleus in response to 0.5- to 2.0- μ M SFN exposure. CONCLUSIONS: The dietary component SFN demonstrates an ability to protect human lens cells against oxidative stress and thus could potentially delay the onset of cataract.